1)

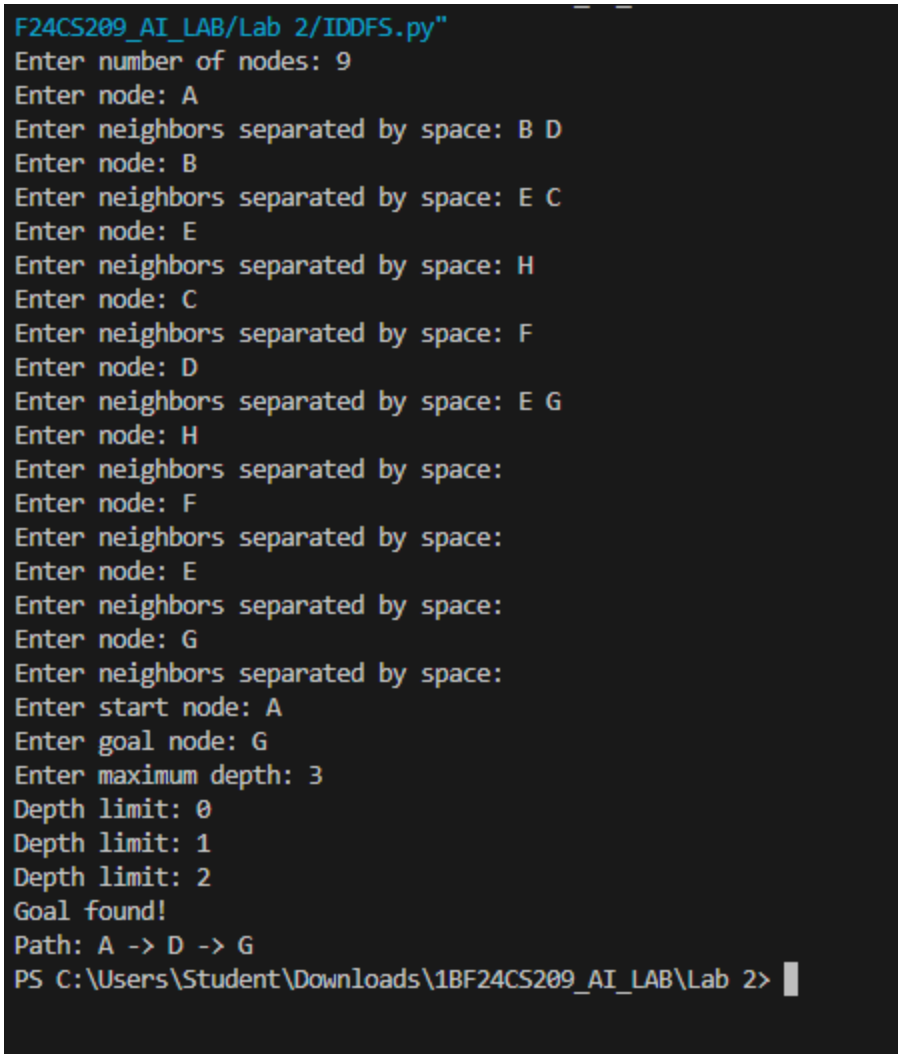
```
F24CS209_AI_LAB/Lab 2/8Puzzle.py"
Enter initial state: 1 2 3 4 0 6 7 5 8
Enter goal state: 1 2 3 4 5 6 7 8 0
Solution found:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

Total moves: 2
PS C:\Users\Student\Downloads\1BF24CS209_AI_LAB\Lab 2>
```

2)


F24CS209_AI_LAB/Lab 2/IDDFS.py"
Enter number of nodes: 9
Enter node: A
Enter neighbors separated by space: B D
Enter node: B
Enter neighbors separated by space: E C
Enter node: E
Enter neighbors separated by space: H
Enter node: C
Enter neighbors separated by space: F
Enter node: D
Enter neighbors separated by space: E G
Enter node: H
Enter neighbors separated by space:
Enter node: F
Enter neighbors separated by space:
Enter node: E
Enter neighbors separated by space:
Enter node: G
Enter neighbors separated by space:
Enter start node: A
Enter goal node: G
Enter maximum depth: 3
Depth limit: 0
Depth limit: 1
Depth limit: 2
Goal found!
Path: A -> D -> G
PS C:\Users\Student\Downloads\1BF24CS209_AI_LAB\Lab 2>

3)

```
y"
Enter number of nodes: 6
Enter node: S
Enter number of neighbors: 2
Enter neighbor and cost: A 1
Enter neighbor and cost: G 10
Enter node: A
Enter number of neighbors: 2
Enter neighbor and cost: B 2
Enter neighbor and cost: C 1
Enter node: B
Enter number of neighbors: 1
Enter neighbor and cost: D 5
Enter node: C
Enter number of neighbors: 2
Enter neighbor and cost: D 3
Enter neighbor and cost: G 4
Enter node: D
Enter number of neighbors: 1
Enter neighbor and cost: G 2
Enter node: G
Enter number of neighbors: 0
Enter heuristic for S: 5
Enter heuristic for A: 3
Enter heuristic for B: 4
Enter heuristic for C: 2
Enter heuristic for D: 6
Enter heuristic for G: 0
Enter start node: S
Enter goal node: G
Optimal path: S -> A -> C -> G
Total cost: 6
PS C:\Users\Student\Downloads\1BF24CS209_AI_LAB\Lab 2>
```

4)

```
ng.py"
Enter number of nodes: 4
Enter node: A
Enter neighbors separated by space: B C
Enter node: B
Enter neighbors separated by space: D
Enter node: C
Enter neighbors separated by space: D
Enter node: D
Enter neighbors separated by space:
Enter heuristic for A: 10
Enter heuristic for B: 6
Enter heuristic for C: 4
Enter heuristic for D: 0
Enter start node: A
Enter goal node: D
Goal found: D
PS C:\Users\Student\Downloads\1BF24CS209_AI_LAB\Lab 2>
```